

BASICS IN LINEAR ALGEBRA AND NUMERICAL METHODS

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General principles, outline

- ▶ 6 classes: 12 lectures.
- ▶ Each lecture = theoretical part (classical lecture) + practical part.
 - ▶ Theoretical part: a collection of knowledge in linear algebra and related topics.
 - ▶ Learning material: this lecture notes.
 - ▶ Practical part: simple practice in Python.
 - ▶ Learning material: Python notebooks.
- ▶ Many notions, knowledge: will be extended in other subjects.
- ▶ Theoretical part: explanations, examples rather than a stiff mathematical discussion.

Lecture 1: an outline

- ▶ What is a matrix?
- ▶ Linear algebra in a nutshell :
 - ▶ vectors, matrices, basic operations
 - ▶ basic operations with vectors: flipping, concatenating, canceling, sorting
- ▶ Matrix and vector manipulations in Python.

Lecture 1: Matrices everywhere - a practical approach

Definition

A matrix B of size $n \times k$ is a table of real numbers with n rows and k columns.

Notations: $B \in \mathbb{R}^{n \times k}$, $B[i, j]$ element in the i th row and j th column.

Several kinds of data in this format:

- ▶ A grayscale picture $P \in \mathbb{R}^{1920 \times 1080}$.
 - ▶ Here $P[i, j]$ is the brightness of pixel point (i, j) .
 - ▶ Indeed, elements are non-negative integers here.
- ▶ Temperature data T of the past 10 years (measured at a fixed point in every hour).
 - ▶ Here $T \in \mathbb{R}^{24 \times 3653}$.
 - ▶ Here $T[i, j]$ is the temperature in the i th hour of the j th day.
 - ▶ Columns corresponds to days, rows to fixed hours.

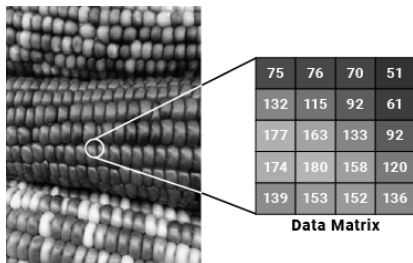


Figure: Matrix representation of a Gray scale picture. Source: https://fr.mathworks.com/help/matlab/creating_plots/image-types.html

Lecture 1: Matrices everywhere (continued)

Several kinds of data in this format (continued):

- ▶ Number of (COVID-19) hospitalized people I in the EU countries in the last 500 days. $\mathbb{R}^{500 \times 27}$.
 - ▶ Here $I[i, j]$ is the number of hospitalized people in the j the country at the i th day.
 - ▶ Again, elements are non-negative integers here.
- ▶ Pressure data P in a horizontal grid for a one-day global weather forecast.
 - ▶ Here $P \in \mathbb{R}^{10^4 \times 5 \cdot 10^5}$.
 - ▶ Here $P[i, j]$ is the pressure in the i th time step (of the numerics) at the j th grid point.
 - ▶ Columns correspond to fixed grid points, rows to fixed time steps.
 - ▶ Rather big: but even more is necessary (more quantities in the equations, also a vertical grid, longer time-span).

Lecture 1: Matrices, operations

- ▶ We need to use operations between matrices.
- ▶ We have to perform them efficiently.
- ▶ We have even to extend this concept to multidimensional arrays.
- ▶ Further practical issues: in which format do we store them, how to read/load them into Python, etc.

Matrix operations may seem a bit abstract, but they make a lot of sense geometrically! (To be continued...)

Lecture 1: Matrix operations I.

Adding of matrices:

- ▶ $A + B$ is defined for matrices of the **same size**.
- ▶ We have to perform it element-wise:

$$A + B[i, j] := A[i, j] + B[i, j]$$

Multiplication of matrices with numbers (scaling):

- ▶ λA is defined for any $\lambda \in \mathbb{R}$ and $A \in \mathbb{R}^{n \times k}$.
- ▶ $\lambda A[i, j] := \lambda \cdot A[i, j]$.

Lecture 1: Matrix operations II. (preparation)

Vectors and their scalar products.

- ▶ $\mathbf{v}$ is a vector of length d consisting of real numbers

Notation: $\mathbf{v} = (v_1, v_2, \dots, v_d) \in \mathbb{R}^d$ (or $\mathbb{R}^{d \times 1}$)

- ▶ $\mathbf{v} \in \mathbb{R}^{d \times 1}$ – column vector; $\mathbf{v} \in \mathbb{R}^{1 \times d}$ – row vector
- ▶ Both of them are indeed, matrices.
- ▶ Python distinguish them (matrix or vector), Matlab does not.
 - ▶ At the same time: row vector = column vector in Python.
- ▶ Scalar product of vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{d \times 1}$:
$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_d v_d.$$
- ▶ Notation (Python): $B[i, :]$ – i th row of the matrix B .

Lecture 1: Matrix operations II.

Multiplication of matrices:

▶ AB is defined for matrices $A \in \mathbb{R}^{n \times k}$ and $B \in \mathbb{R}^{k \times m}$.

▶ $AB \in \mathbb{R}^{n \times m}$:

$$AB[i, j] := A[i, :] \cdot B[:, j]' = A[i, :] \cdot B[:, j]^T$$

▶ Indeed, both of them are row vectors of size k .

▶ Important special case: $A \in \mathbb{R}^{n \times k}$, $\mathbf{v} \in \mathbb{R}^{k \times 1}$.

▶ “Matrix times a column vector is a column vector”.

▶ Computational costs: each element of the matrix should be multiplied with a vector element (exactly once).

▶ Summarized: $n \cdot k$ multiplications (and $n \cdot (k - 1)$ additions).

Lecture 1: Matrix operations III.

- ▶ Transposing: “rows $\leftrightarrow$ columns”; formally: $B^T[i, j] := B[j, i]$.
 - ▶ If $B \in \mathbb{R}^{n \times k}$ then $B^T \in \mathbb{R}^{k \times n}$
 - ▶ If $B^T = B$, it is called *symmetric*.
- ▶ An important matrix: $I \in \mathbb{R}^{n \times n}$ – identity matrix, unit matrix.
 - ▶ $I[1, 1] = I[2, 2] = \dots I[d, d] = 1$ (called main diagonal)
 - ▶ $I[j, k] = 0$ for $j \neq k$ (outside of the main diagonal)
- ▶ For all $A \in \mathbb{R}^{n \times n}$: $AI = IA = A$.
 - ▶ Just like 1 for real numbers.
- ▶ For $A \in \mathbb{R}^{n \times n}$: A^{-1} is the matrix (if exists!) satisfying $AA^{-1} = A^{-1}A = I$.

Lecture 1: vector operations

A number of further useful vector-operations:

- ▶ flipping,
- ▶ concatenating,
- ▶ cancelling some elements,
- ▶ sorting.

Built-in functions for them in Matlab and Python.

- ▶ More details in the practical part.

Lecture 1: preliminary remarks for implementation in Python

Be careful:

- ▶ multiplication, taking power, taking built-in functions are defined between matrices or vectors *of the same size and componentwise*.
- ▶ Also, numbering of elements, rows begin with *zero*.
- ▶ Number of the last item (row, column) is -1,
 - ▶ this can be handy.

Lecture 2: an outline

- ▶ Short additional material:
 - ▶ Representation, partitions,
 - ▶ equivalence relations.
- ▶ Geometrical interpretation of matrices:
 - ▶ Subspaces,
 - ▶ range of matrices,
 - ▶ projections.
- ▶ Matrix manipulations in Python – continued.

Lecture 2: Partition of sets: motivation

- ▶ A basic example:
 - ▶ A number of pictures are given,
 - ▶ some of them are on dogs, some on cats and some on houses.
 - ▶ It is inherently partitioned (categorized).
 - ▶ The computer should find this partition.
- ▶ Another example:
 - ▶ Human DNS-fractures are found in an archaeological study.
 - ▶ Decide whether this belongs to a men or women.
 - ▶ Decide whether this belongs to a child or to an adult.
 - ▶ Decide to which nation (out of six) can this belong.

Lecture 2: Partition of sets: mathematical modeling

- ▶ In a general setting:
 - ▶ A set H is given.
 - ▶ Abstraction of all samples or pictures.
 - ▶ There is an *equivalence relation* $\sim$ between (some of) them.
 - ▶ Abstraction of "belong to the same class/partition".
 - ▶ This should be
 - ▶ **reflexive:** $h \sim h$ for all $h \in H$.
 - ▶ **symmetric:** $a \sim b \Rightarrow b \sim a$ for all $a, b \in H$.
 - ▶ **transitive:** $a \sim b \ \& \ b \sim c \Rightarrow a \sim c$ for all $a, b, c \in H$.

Lecture 2: Partition of sets II.

- ▶ Using such a relation, we obtain a non-intersecting partition of H
 - ▶ One part (equivalence class)- (sub)set of equivalent elements.
 - ▶ This is well-defined (also for any infinite sets).
- ▶ Mathematical example:
 - ▶ $H = \mathbb{R}^2$, $\mathbf{v} \sim \mathbf{u} \Leftrightarrow v_1 = u_1$
 - ▶ Equivalent vectors have the same first coordinate.
 - ▶ Equivalence classes: vertical lines.
- ▶ Another example:
 - ▶ $H =$ points of a given network, $x_1 \sim x_2 \Leftrightarrow$ there is a path from x_1 to x_2 .
 - ▶ Equivalent points can be reach from each other.
 - ▶ Equivalence classes: connected subsets.

Lecture 2: Partition of sets II. - an example, additional remarks

- ▶ Again: we take a set of pictures: $\{P_j : j \in J\}$
 - ▶ More precisely, we have to identify the object $Q_j \in P_j$ in the picture.
- ▶ We usually assume:
 - ▶ For all $j \in J$: (rotation of Q_j) $\in P_k \Rightarrow P_k \sim P_j$.
 - ▶ For all $j \in J$: (any shift of Q_j) $\in P_k \Rightarrow P_k \sim P_j$.
 - ▶ For all $j \in J$: (any mirror image of Q_j) $\in P_k \Rightarrow P_k \sim P_j$.
- ▶ Is it also OK?
 - ▶ P_j and P_k coincide with 99% $\Rightarrow P_j \sim P_k$
 - ▶ No, after a number of steps, transitivity is lost.

Lecture 2: Partition of sets III.

- ▶ Real-life situations:
 - ▶ A meaningful partition should be given: dogs, cats, houses on pictures.
 - ▶ Possibly we know the number or nature of the partition: (samples on infected or non-infected patients).
 - ▶ Possibly we do not know classes – we have to identify some properties to find them.
 - ▶ Example: classification of cancer types (based on patterns in their DNA).
 - ▶ An appropriate AI-based algorithm can do it better than experts.

Lecture 2: Back to linear algebra – linear mappings

- ▶ A linear mapping $B : \mathbb{R}^k \rightarrow \mathbb{R}^n$:
 - ▶ For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^k$: $B(\mathbf{u} + \mathbf{v}) = B\mathbf{u} + B\mathbf{v}$,
 - ▶ for all $\lambda \in \mathbb{R}$: $B(\lambda\mathbf{u}) = \lambda B\mathbf{u}$.
- ▶ Main example: a matrix multiplication with $B \in \mathbb{R}^{n \times k}$.
- ▶ “Visible” examples: reflection, rotation.
 - ▶ Shifting in $\mathbb{R}^3$ is also linear?
 - No, be careful; if A is a linear map, $A\mathbf{0} = \mathbf{0}$ should be valid.

Lecture 2: Linear mappings and matrices

- ▶ From this point of view: a matrix of size $n \times k$ is a table representing a linear mapping $\mathbb{R}^k \rightarrow \mathbb{R}^n$.
- ▶ Algorithm to construct it: j th column is the vector of the image of the unit vector $\mathbf{e}_j$.
- ▶ Example 1: Rotation with $+90^\circ$ in $\mathbb{R}^2$.
 - ▶ This should be a matrix of size 2×2 .
 - ▶ First column should be the image of $(1, 0)$, which is $(0, 1)^T$.
 - ▶ Second column should be the image of $(0, 1)$, which is $(-1, 0)^T$.
 - ▶ Summarized, we got: $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.
- ▶ Same notation for the matrix and the linear mapping.

Lecture 2: Linear mappings and matrices: change the basis

- ▶ Sometimes, matrices have a simple structure if they are given in another basis.
- ▶ Instead of taking the unit vector $\mathbf{e}_j$ for all j above, one should choose a *basis* $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$ (and possibly also in $\mathbb{R}^n$) to have a simpler matrix.
- ▶ Example: Reflection to the line $y = \frac{\sqrt{3}}{2}x$ in $\mathbb{R}^2$.
 - ▶ Using the conventional basis:
 - ▶ First column should be the image of $(1, 0)$, which is $(\frac{1}{7}, \frac{4\sqrt{3}}{7})^T$.
 - ▶ Second column should be the image of $(0, 1)$, which is $(\frac{4\sqrt{3}}{7}, -\frac{1}{7})^T$.
 - ▶ Summarized, we got: $B = \begin{pmatrix} \frac{1}{7} & \frac{4\sqrt{3}}{7} \\ \frac{4\sqrt{3}}{7} & -\frac{1}{7} \end{pmatrix}$.

Lecture 2: change the basis - example continued

- ▶ Take the following basis:
- ▶ $\mathbf{b}_1 = (1, \frac{\sqrt{3}}{2})$, $\mathbf{b}_2 = (-\frac{\sqrt{3}}{2}, 1)$
- ▶ First column should be the image of $\mathbf{b}_1$ is itself: $(1, 0)^T$ in the basis $(\mathbf{b}_1, \mathbf{b}_2)$.
- ▶ Second column should be the image of $\mathbf{b}_2 = (-\frac{\sqrt{3}}{2}, 1)$, which is the opposite of itself: $(0, -1)^T$ in the basis $(\mathbf{b}_1, \mathbf{b}_2)$.
- ▶ Summarized, we got: $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
- ▶ This is really a more nice form; we extend this idea later on.

Lecture 2: Further important linear mappings

► Projections

- Example: projection P to the horizontal plane within $\mathbb{R}^3$.

- Then $(1, 0, 0) \xrightarrow{P} (1, 0, 0)$, $(0, 1, 0) \xrightarrow{P} (0, 1, 0)$ and finally, $(0, 0, 1) \xrightarrow{P} (0, 0, 0)$.

- In this way: $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

- But we could represent it using a smaller matrix:

- Then $(1, 0, 0) \xrightarrow{P_2} (1, 0)$, $(0, 1, 0) \xrightarrow{P_2} (0, 1)$ and finally, $(0, 0, 1) \xrightarrow{P_2} (0, 0)$.

- In this way: $P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

- We will generalize this observation: if a matrix consist of just a few linearly independent columns, we can represent it using less size/entries/memory.

Lecture 2: Linear mappings and matrices

- ▶ Matrix product: composition of the corresponding linear mappings.

- ▶ AB makes only sense if $B : \mathbb{R}^n \xrightarrow{B} \mathbb{R}^k \xrightarrow{A} \mathbb{R}^m$:

- ▶ Also, we have: $AB \in \mathbb{R}^{n \times m}$.

- ▶ For square matrices power makes sense: $A^k = \underbrace{AA \cdots A}_{k \text{ times}}$.

- ▶ Unit matrix $I \in \mathbb{R}^{k \times k}$ corresponds to the identical mapping:

$$I\mathbf{v} = \mathbf{v} \text{ for all } \mathbf{v} \in \mathbb{R}^k.$$

- ▶ Matrix inverse corresponds to the inverse mapping:

$$\mathbf{v} \xrightarrow{B} B\mathbf{v} \xrightarrow{B^{-1}} \mathbf{v}.$$

- ▶ Do not compute this in the practice (very slow).